Why Normalization Does Not Force s_1 and s_2 to Unity

Question. The rectangular forward initializes two per-experiment coefficients at one and uses fixed probe and detector scaling. Does that construction force $s_1, s_2 \approx 1$, leaving them only a small modulation such as 0.9–1.1?

Short answer. No. The fixed factors cancel known preprocessing and choose detector/probe units; they do not normalize the unknown object coefficients. Unity is an initialization (or a plausible scale under a well-matched gauge), not an invariant or bound.

Symbol	Meaning
P_{phys}	complex probe in the chosen acquisition gauge
q	fixed probe normalization, $P_{\text{train}} = qP_{\text{phys}}$
d	fixed legacy loader intensity-normalization metadata
k	derived exit-wave amplitude factor passed to the forward as `scale`
a, b	learned dimensionless real/imaginary decoder textures
s_1, s_2	per-experiment real coefficients multiplying those textures

What the fixed scales actually cancel

The autograd rectangular forward computes

$$\psi = k\left[s_1(P_{\text{train}}a) + i\, s_2(P_{\text{train}}b)\right], \qquad I_{\text{pred}} = \sum_p \left|\mathcal{F}\{\psi_p\}\right|^2. \tag{1}$$

Here k is a real scalar; it is not the complex probe, which is the separate factor P_{train}.

In the loader-normalized legacy implementation, the notation maps directly to

$$q \longleftrightarrow \texttt{batch[2]},$$
$$d \longleftrightarrow \texttt{fields['physics\_scaling\_constant']}.$$

The loss code constructs

$$\texttt{exit\_wave\_scale} \equiv k = (q^2 d + 10^{-9})^{-1/2} \tag{2}$$

under the variable name `output_scale`; the rectangular forward then receives that value through its argument named `scale`.

For the current `ci_intensity_v2` count-intensity path,

$$k = q^{-1}, \qquad kP_{\text{train}} = q^{-1}(qP_{\text{phys}}) = P_{\text{phys}}. \tag{3}$$

Thus the output factor exactly undoes training-probe normalization. There is no `physics_scale` in this CI path; measured intensities remain counts.

For the explicitly loader-normalized `legacy_v1` rectangular path,

$$k = (q^2 d + 10^{-9})^{-1/2} \approx \frac{1}{q\sqrt{d}}, \qquad kP_{\text{train}} \approx \frac{P_{\text{phys}}}{\sqrt{d}}. \tag{4}$$

This statement is specific to that adapter; other legacy adapters, including dictionary parity, need not supply the same factors. In either path, the fixed terms remove known representation choices. They leave the unknown object

$$O = s_1 a + i\, s_2 b \tag{5}$$

inside the physical forward. Nothing in the cancellation sets either coefficient to one.

Why the coefficients are not identified near one

First, s_1 and s_2 describe real and imaginary object contrast, not merely residual calibration. Those contrasts need not be equal or have unit scale.

Second, the model has an internal factorization freedom. Wherever the rescaled texture remains in the decoder's representable range,

$$s_1 a = (cs_1)(a/c), \qquad s_2 b = (rs_2)(b/r). \tag{6}$$

The detector loss observes the products, not each factor separately. A bounded decoder limits this freedom but does not fix a texture norm, so it does not select $s_1 = s_2 = 1$.

Third, ptychography has a distinct upstream acquisition gauge:

$$P_{\mathrm{phys}} O = (\gamma P_{\mathrm{phys}})(O/\gamma). \tag{7}$$

Storing a physical probe chooses one such decomposition; diffraction alone does not establish absolute object units. This acquisition gauge is separate from the train-time decoder factorization above.

Initialization versus construction

The module constructor creates $s_1 = s_2 = 1$. With `ones` initialization, training starts there. With `dose_closure`, both are instead set to a data-derived common gauge $g > 0$ so a unit-object prediction starts near the measured dose; g need not equal one. During autograd training the coefficients are ordinary unconstrained parameters: there is no range, clamp, or regularizer keeping them near unity.

Correct normalization can therefore make order-unity values plausible and improve conditioning. It cannot guarantee them. Values far from one may reflect legitimate object contrast, decoder rescaling, the chosen probe–object gauge, or inconsistent scaling metadata.

Assessment of the quoted interpretation. The derived output factor is indeed passed as the forward argument named `scale`, and fixed factors do absorb known preprocessing. However, that `scale` is not the complex probe, and "s_1, s_2 stay around one by construction" is false. At most, near-one behavior is an empirical expectation under a compatible gauge and training trajectory.